```
> restart;# The spin orf proton & electron
```

```
> R := 3.9228·10^22; mp := 1.672621923·10^-27; hbar := 6.62607015/(2·Pi)·10^-34; rp := 1.2·10^-15; m := 9.10938356·10^-31; c := 299792458; ec := 1.602176·10^-19; h := 6.62607015·10^-34; B1 := 1.702112406; B2 := 3.81·10^14; ϵ := 8.854187813 × 10^-12; rb := 5.292410932 × 10^-11; re := 2.82·10^-15; hm := 5.272227549 × 10^6·h; r0 := 4.2·10^-15; hme := 2.180503182 × 10^-8; E0 := 5.747437648 × 10^-19; 1/ec;
#B2 is the proton's inner magnetic field B2 = 3.81□1014 Tesla
link: https://www.sciencedirect.com/science/article/pii/S2211379718316176
```

$$R := 3.922800000 \times 10^{22}$$

$$mp := 1.672621923 \times 10^{-27}$$

$$\hbar := 1.054571818 \times 10^{-34}$$

$$rp := 1.200000000 \times 10^{-15}$$

$$m := 9.109383560 \times 10^{-31}$$

$$c := 299792458$$

$$ec := 1.602176000 \times 10^{-19}$$

$$h := 6.626070150 \times 10^{-34}$$

$$B1 := 1.702112406$$

$$B2 := 3.810000000 \times 10^{14}$$

$$\epsilon := 8.854187813 \times 10^{-12}$$

$$rb := 5.292410932 \times 10^{-11}$$

$$re := 2.820000000 \times 10^{-15}$$

$$hm := 3.493414959 \times 10^{-27}$$

$$r0 := 4.200000000 \times 10^{-15}$$

$$hme := 2.180503182 \times 10^{-8}$$

$$E0 := 5.747437648 \times 10^{-19}$$

$$6.241511544 \times 10^{18} \quad \textbf{(1)}$$

```
> ωe := 2·Pi·(5·ec·B2·h/(re^2·m))^0.5;
#The angular precession of electron in "trembling" precession per second
```

$$\omega e := 1.049824929 \times 10^{12} \quad \textbf{(2)}$$

```
> ωp := 2 Pi·(5·ec·B2·h/(rp^2·mp))^0.5;
#The angular precession of proton per second
```

$$\omega p := 5.757452262 \times 10^{10} \quad \textbf{(3)}$$

```
> Ee := B2·ec·hm/5;
```

(4)

$$Ee := 4.264963992 \times 10^{-32} \quad (4)$$

```
> ω2 := (5·h·ec·B1/(m·re^2))^0.5;
  #the rize of angular precession of electron  due to magnetic
  field of nucleus;
```

$$\omega 2 := 11167.82613 \quad (5)$$

```
> Er := ec^2/(2·4·Pi·ϵ·rb);#The Rydberg Energy in Joule
```

$$Er := 2.179607514 \times 10^{-18} \quad (6)$$

```
> E := 3.59;#The Energy spin in eV
```

$$E := 3.59 \quad (7)$$

```
> E0 := E/13.60567675·Er;#The spin Energy Joule
```

$$E0 := 5.751122211 \times 10^{-19} \quad (8)$$

```
> E0·mp/(ec·B2·h); E0·m/(ec·B2·h)# [J·Kg·A^-1·s^-1Kg^-1·s^2·A·J^-1·s^-1]= [0]
```

$$2.378259029 \times 10^{-8}$$

$$1.295240329 \times 10^{-11} \quad (9)$$

```
> B1 := E0·h/(ec·hm·4·10^-7);
  # The magnetic field due to the precession of the proton
  where E0 is the spin energy asociated to E0=hν =
  5.751122211×10^-19J
```

$$B1 := 1.702112406 \quad (10)$$

```
> (5·ec·B2·h/(rp^2·mp))^0.5; 1/%
```

$$9.163269869 \times 10^{9}$$

$$1.091313488 \times 10^{-10} \quad (11)$$

```
> 2·Pi·(5·ec·B2·h/(rp^2·mp))^0.5;
```

$$5.757452262 \times 10^{10} \quad (12)$$

```
> 2·Pi·(5·ec·B1·h/(re^2·m))^0.5;
```

$$70169.52106 \quad (13)$$

```
> Fspin := 4·Pi^2·9.146 ·10^-10·ωp·rp ;
  # The aproximation of atraction forces due to precession of
  nucleons
```

$$Fspin := 2.494609234 \times 10^{-12} \quad (14)$$

```
> F1 := ec·(2·E0/mp)^0.5 ·B2
```

$$F1 := 1.600765078 \quad (15)$$

```
> F1 := ec·(ωp·rp)^0.5·B2;
```

$$F1 := 5.073889365 \times 10^{-7} \tag{16}$$

```
> 3.81·10^14·5.79·10^-9;
```

$$2.205990000 \times 10^{6} \tag{17}$$

```
> B2 := 3.81·10^14; B2·0.25·1.05457·10^-34·9.57865·10^7·6.2415·10^18;# Sn 117
```

$$B2 := 3.810000000 \times 10^{14}$$

$$6.005285736 \times 10^{6} \tag{18}$$

```
> B2·0.25·1.05457·10^-34·10.01926·10^7·6.2415·10^18;# Sn 119
```

$$6.281523930 \times 10^{6} \tag{19}$$

```
> 0.25·4.815·10^7·B2·6.2415·10^18·1.05457·10^-34 ;#Hg 199
```

$$3.018739679 \times 10^{6} \tag{20}$$

```
> 1.777·10^7·0.75·B2·6.2415·10^18·1.05457·10^-34;#Hg 201
```

$$3.342243245 \times 10^{6} \tag{21}$$

```
> E0/hm; c·hm/E0 ; ec/(m·B2)·4·10^-7; ec/m; ec/mp;
    # thus 180.91Mhz and  wavelenght =1.657128548m
```

$$1.646275143 \times 10^{8}$$

$$1.821034954$$

$$1.846529493 \times 10^{-10}$$

$$1.758819342 \times 10^{11}$$

$$9.578829369 \times 10^{7} \tag{22}$$

```
> restart;#GyroMag
```

```
> mp := 1.672621923·10^-27; hbar := 6.62607015/(2·Pi)·10^-34; B2 := 3.81·10^14;
    ec := 1.602176·10^-19; 1/ec; NA := 6.021760000×10^23 ;
    #Avogadro number[g/mol]
```

$$mp := 1.672621923 \times 10^{-27}$$

$$\hbar := 1.054571818 \times 10^{-34}$$

$$B2 := 3.810000000 \times 10^{14}$$

$$ec := 1.602176000 \times 10^{-19}$$

$$6.241511544 \times 10^{18}$$

$$NA := 6.021760000 \times 10^{23} \tag{23}$$

```
> γ0 := 9.57865·10^7; I0 := 1/2; γ0 := 10.01926·10^7; I0 := 1/2; γ0 := 4.81519·10^7;
    I0 := 1/2; γ0 := -1.77748·10^7; I0 := 3/2; γ0 := 5.64661·10^7; I0 := 1/2;
```

$$\gamma 0 := 9.578650000 \times 10^{7}$$

$$I0 := \frac{1}{2}$$

$$\gamma 0 := 1.001926000 \times 10^{8}$$

$$I0 := \frac{1}{2}$$

$$\gamma 0 := 4.815190000 \times 10^{7}$$

$$I0 := \frac{1}{2}$$

$$\gamma 0 := -1.777480000 \times 10^{7}$$

$$I0 := \frac{3}{2}$$

$$\gamma 0 := 5.646610000 \times 10^{7}$$

$$I0 := \frac{1}{2} \quad \textbf{(24)}$$

```
> E = hbar·γ0·I0·(1 − I0)·B2/ec;#EnergySn117=6.620023680×10^6; Sn119
  = 6.924539308 × 10^6; Hg 199 = 3.327887733 × 10^6; Hg201 = 3.685371015
  × 10^6; Pb207 = 3.902501074 × 10^6
```

$$E = 3.540125974 \times 10^{6} \quad \textbf{(25)}$$

```
> P := 2·ec/(2.378259029 × 10^-8);
  # The energy of 1-2 eV and the interaction time with Hg
  nucleus gives 1.35·10^-11 Watts per atom
```

$$P := 1.347351975 \times 10^{-11} \quad \textbf{(26)}$$

```
> Pgram := P·NA/200# Tha maximum power 4·10^10 Watts per time unit
  ·(40 GW) contained in 1 gram of  Hg
```

$$Pgram := 4.056715114 \times 10^{10} \quad \textbf{(27)}$$

```
>
```